



Supplementary Materials for

Wafer-scale monolayer MoS₂ film integration for stable, efficient perovskite solar cells

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Science **387**, 186 (2024)
DOI: 10.1126/science.ado2351

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Tables S1 and S2

Other Supplementary Material for this manuscript includes the following:

Movie S1

Materials and Methods

Materials

All reagents were used as received without further purification. Isopropanol (IPA, 99.5% anhydrous), Chlorobenzene (CB, 99.8% anhydrous), PbI_2 (99.999%), N,N-dimethylformamide (DMF, 99.8% anhydrous), and Dimethyl sulfoxide (DMSO, $\geq 99.9\%$ anhydrous) were purchased from Sigma-Aldrich. Poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA), Formamidinium iodide (FAI), Methylammonium chloride (MACl), C_{60} and Bathocuproine (BCP) were purchased from Xi'an Polymer Light Technology. Tetrakis(dimethylamino)tin(IV) (TDMASn , 99.9999%) was purchased from Beijing Timestorch. 2,3,5,6-Tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ) were purchased from Luminescence Technology Corp. (Lumtec). Indium tin oxide (ITO) glass substrates were purchased from Beijing HuaMin New Materials Technology Co., Ltd. Au, Ag, Cu and ITO target were purchased from ZhongNuo Advanced Material (Beijing) Technology Co., Ltd.

Methods

Synthesis of monolayer MoS_2 film. The monolayer MoS_2 films were prepared via the face-to-face metal-precursor feeding route following previous work. Briefly, a rectangular piece of Mo foil was folded into a bridge and placed over a clean soda-lime glass. The Mo foil and glass were placed in a graphite boat and loaded into the CVD chamber. Another quartz boat containing sulfur powder was placed 40 cm upstream from the glass substrate. Then, Ar (50 sccm) and O_2 (6 sccm) gas flows were introduced into the system to create a stable growing atmosphere. The sulfur zone and the substrate zone were heated to 100 and 720 $^\circ\text{C}$ within 35 min, respectively. Monolayer MoS_2 film was obtained on the glass substrate after 8 min growth, and then cooled to room temperature naturally with the furnace.

Transfer of monolayer MoS_2 film. First, the edge of as-grown glass/ MoS_2 was scratched with a knife to provide a path for water penetration. Then, thermal release tape (TRT) was stuck on the MoS_2 film by vacuum lamination and the glass/ MoS_2 /TRT was subsequently immersed in deionized water. Due to the difference in surface energy between MoS_2 and glass substrate, water naturally penetrated into the interface between MoS_2 and glass substrate. After MoS_2 /TRT was completely delaminated from the glass substrate, it was picked up and then blow-dried with N_2 flow. Next, the dried MoS_2 /TRT was pressed onto a target substrate by vacuum lamination. Finally, the target substrate with MoS_2 /TRT was heated to 100 $^\circ\text{C}$ on a hot-plate. The TRT lost adhesion at high temperature and automatically peeled off, leaving the MoS_2 film onto the target substrate (Movie S1).

Synthesis and transfer of monolayer WS_2 film. The monolayer WS_2 films were prepared via a facile APCVD process. The quartz boat containing sulfur powder was placed upstream and the heating temperature was set at 200 $^\circ\text{C}$. The graphite boat containing WO_3 powder was placed in the middle of the furnace, and a c-plane sapphire substrate was placed on the graphite boat with the surface facing down. The distance between S and WO_3 precursors was 45 cm. The furnace was heated to 850 $^\circ\text{C}$ in 45 min and then kept for 20 min to obtain the continuous monolayer WS_2 film. The whole growth process was under the Ar (50 sccm) and H_2 (3 sccm) gas flows, and the furnace was quickly cooled down to the room temperature under the gas of Ar. The WS_2 film was transferred following the same procedure as MoS_2 film.

Device fabrication. ITO glass was sequentially cleaned using detergent, ultra-pure water, and isopropanol for 20 min, respectively. Before use, the glass/ITO substrate was treated with O_2 -plasma for 5 minutes. The PTAA solution was prepared with PTAA (2 mg), F4-TCNQ (0.2 mg)

and CB (1 mL). The as-prepared PTAA solution was spin-coated on ITO substrate at 4,000 rpm for 30 s, and the substrate was subsequently annealed on a hot plate at 100 °C for 10 min. After cooling to room temperature, monolayer MoS₂ film was transferred to the PTAA surface. The 1.5 M PbI₂ precursor solution was prepared in a mixed solvent of DMF and DMSO with a volume ratio of 9:1. The PbI₂ film was spin-coated onto PTAA/MoS₂ at 2,000 rpm for 30 s, then annealed at 70 °C for 1 min. For perovskite film deposition, a solution of FAI: MACl (90 mg: 10 mg in 1 ml IPA) was spin-coated onto the PbI₂ film at 2,700 rpm for 10 s. The substrate was then immediately transferred on a hot plate and heated at 150 °C for 10 min. After cooling down to room temperature, the bimolecular passivator with IPA/DMF mixed solvent (200:1, v:v) was spin-coated onto the perovskite surface at 4,000 rpm for 30 s, followed by thermal annealing at 100 °C for 5 min. Then, monolayer MoS₂ film was transferred to the perovskite surface. Finally, C₆₀ (20 nm), BCP (6 nm) and Ag (or Cu, 200 nm) were sequentially deposited by thermal evaporation at the rates of 0.2 Å/s, 0.2 Å/s and 1.0 Å/s, respectively. For FA_{0.93}Cs_{0.07}PbI₃-based solar cells, the PbI₂ precursor solution was prepared in a mixed solvent of DMF and DMSO with a volume ratio of 9:1, containing PbI₂ (1.5 M), CsI (0.105 M), and other fabrication process remained unchanged.

For n-i-p solar cells, the SnO₂ colloidal solution (2.5 wt%) was spin-coated on ITO substrate at 4,000 rpm for 30 s, and annealed on a hot plate at 150 °C for 30 min in ambient air. After cooling to room temperature, monolayer MoS₂ film was transferred to the SnO₂ surface. After that, PbI₂ precursor solution (1.5 M in a mixed solvent of DMF and DMSO with a volume ratio of 900:100) was spin-coated onto SnO₂/MoS₂ at 2,000 rpm for 30 s, then annealed at 70 °C for 1 min. For perovskite film deposition, a solution of FAI: MACl (90 mg: 10 mg in 1 ml IPA) was spin-coated onto the PbI₂ film at 2,700 rpm for 10 s. The substrate was then immediately transferred on a hot plate and heated at 150 °C for 10 min. After the perovskite deposition, monolayer MoS₂ film was transferred to the perovskite surface. Then the hole-transport layer was subsequently deposited on top of the perovskite/MoS₂ by spin coating at 3,000 rpm for 30 s using a spiro-OMeTAD/CB solution, which consisted of spiro-OMeTAD (72.3 mg), tBP (30 µL), Li-TFSI/ACN (35 µL, 260 mg/mL) and CB (1 mL). Finally, 100 nm Au was deposited by thermal evaporation.

The minimodules with the architecture of ITO/PTAA/(MoS₂)/FAPbI₃/(MoS₂)/C₆₀/SnO₂/Cu were fabricated on the pre-patterned large ITO glass substrates (4 cm × 4 cm, P1 width 50 µm) following the same procedure as the p-i-n solar cells. The fabricated modules have 5 sub-cells, and each sub-cell has a width of 6 mm and a length of 32 mm. The widths of P2 and P3 were 100 and 70 µm, respectively. The total scribing line width was 0.31 mm, giving a geometric fill factor (GFF) of 94.83%.

Characterization. Scanning electron microscope (SEM) images were measured using Hitachi S4800 field-emission scanning electron microscopy. Atomic force microscope (AFM) was conducted using Bruker (Dimension Icon). The UV-visible absorption spectra were measured by Hitachi UH4150 spectrophotometer. Raman spectroscopy was measured by LabRAM HR-800 (HORIBA) with an excitation wavelength of 514 nm. X-ray diffraction (XRD) patterns were recorded by Rigaku D/Max 2,200 with Cu K α as the X-ray source. The UV-visible absorption spectra were measured by Hitachi UH4150 spectrophotometer. PL spectra and TRPL spectra were obtained by FLS1000 (Edinburgh Instruments Ltd) with an excitation at 470 nm. X-ray photoelectron spectroscopy (XPS) was performed by PHI 5000 VersaProbe III with a monochromatic Al K α X-ray source. Ultraviolet photoelectron spectroscopy (UPS) was performed by PHI 5000 VersaProbe III with He I source (21.22 eV). Raman spectra were measured through a micro-Raman spectrometer (Horiba XploRA PLUS) with an excitation laser of the wavelength of 532 nm at room temperature. The *J-V* measurement was performed via the solar simulator (SS-

C10-3A, Enlitech) along with AM 1.5G spectra whose intensity was calibrated by the certified standard silicon solar cell (SRC-2020, Enlitech, 2019/02/11) at 100 mW/cm². The *J-V* curves were obtained by scanning along the reverse and forward scan direction with the scanning speed of ~70 mV/s. For stabilized power output measurements, cells were fixed at the maximum power point (MPP) voltage (determined from *J-V* sweeps) and the current output was tracked over time. The active area was determined by the aperture shade mask (0.0735 cm²) placed in front of the solar cell. We sent the best solar cell to an accredited laboratory (the PV Metrology Lab of National Institute of Metrology, China) for certification. The external quantum efficiency (EQE) data were obtained by using the solar-cell spectral-response measurement system (QE-R, Enlitech), and a calibrated silicon diode with a known spectral response was used as a reference. Time-of-flight secondary ion mass spectroscopy (ToF-SIMS) measurements were conducted with an ION-TOF V spectrometer. Bi⁺ beam (25 keV) was used as the primary beam to detect the samples and the sputter etching was performed using a Cs⁺ beam (3 keV) to obtain the desired depth profile. The transient photovoltage/photocurrent (TPV/TPC) decay measurements were obtained on Molex 180081-4320 simulating one sun working condition.

Encapsulation and stability testing. For the stability test, 10 nm layer of SnO₂ and 100 nm layer of ITO were deposited on top of the C₆₀ sequentially, by atomic layer and sputtering depositions, respectively, to replace the BCP layer. Specifically, 10 nm SnO₂ was deposited by atomic layer deposition (ALD) with N₂ as the gas carrier. In a typical ALD deposition, the substrate temperature was maintained at 85 °C, with the TDMASn precursor source at 65 °C and the water source at room temperature. In one whole cycle, the dosing and purging time for TDMASn is 0.05 and 6 s, and for H₂O is 0.02 and 6 s, respectively. The total number of cycles was 100. After ALD-SnO₂ deposition, 100 nm ITO was sputtered at 100 W power under an Ar pressure of 0.5 Pa.

Encapsulation was done by vacuum laminating. First, wires were connected to the electrodes of the cells. Then, the cell was encapsulated in two glass sheets with polyolefin elastomer (POE) encapsulant and edge sealing via polyisobutylene (PIB) rubber. Finally, the stack was vacuum laminated in an industrial laminator at 120 °C for 15 min. For the damp-heat tests, encapsulated cells were aged in a damp-heat chamber at constant temperature and relative humidity levels (85 °C and 85%), and measured periodically by *J-V* scans after around 30 min cooling of cells. The operational stability tests were performed by tracking MPPs of unencapsulated cells under 1-Sun illumination in nitrogen at room temperature. The high-temperature operational stability tests were performed by tracking MPPs of encapsulated cells under AM 1.5G illumination at 85 °C in the ambient atmosphere. For thermal cycling tests, encapsulated devices were placed in the environment test chamber, with the temperature cycling between -40 °C and +85 °C. The temperature change rate between -40 °C and +85 °C was set to 100 °C/h, and the temperature was stable for 15 min at the temperature point of -40 °C and +85 °C, respectively.

DFT calculation. The density functional theory (DFT) calculation based on the projector-augmented wave (PAW) method as implemented in the Vienna ab initio simulation package (VASP). The Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) was employed to describe the exchange-correlation interaction. The Grimme potential (D2) was adopted to correct the Van der Waals interaction. The climbing image nudged elastic band (CI-NEB) method based on the VTST tools was used to calculate the transformation barrier between α -FAPbI₃ into δ -FAPbI₃. The cutoff energy was set to be 400 eV. The Monkhorst-Pack k-point mesh was sampled with a separation of about 0.015 Å⁻¹ in the Brillouin zone. All structures were relaxed until the residual force on each atom less than 0.01 eV/Å. The self-consistent convergence accuracy was set at 10⁻⁵ eV/atom in the structural

calculation, and 10^{-7} eV/atom in the CI-NEB calculation. For the MoS₂/perovskite interfaces, they were composed of 3×2 supercell of FAPbI₃(001) surface and 6×4 supercell of MoS₂(001) surface. Two typical FAPbI₃(001) surfaces were considered: one was the PbI termination and the other was the FAI termination. The average lattice mismatches of MoS₂/perovskite interfaces are less than 1.30%. A vacuum region more than 15 Å in the z-direction in conjunction with the dipole correction to avoid the fictitious interaction with its periodic images. Three kinds of point defects including I vacancies (V_I), I interstitials (I_i) and FA vacancies (V_{FA}) were introduced into the MoS₂/perovskite interfaces with FAI terminations, and V_I , I_i and Pb vacancies (V_{Pb}) were introduced into the MoS₂/perovskite interfaces with PbI terminations.

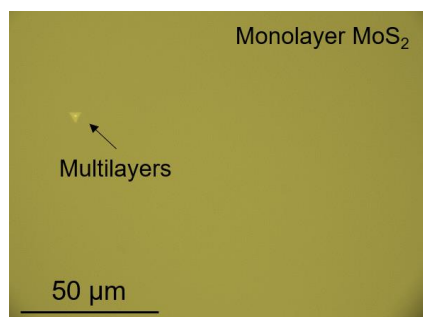


Fig. S1. Typical OM image of the MoS₂ film.

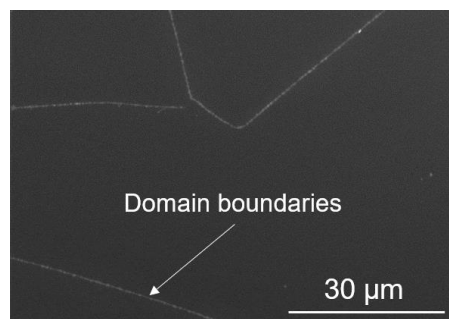


Fig. S2. Typical SEM image of the MoS₂ film.

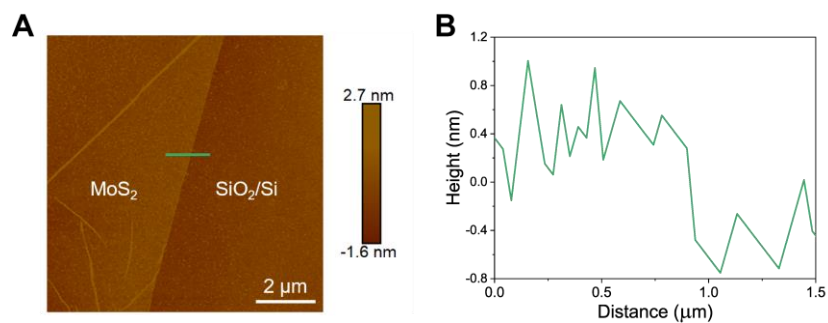


Fig. S3. (A and B) AFM topography image (A) of the MoS₂ film transferred onto SiO₂/Si and its corresponding height profile (B).

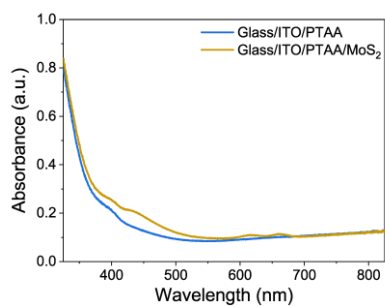


Fig. S4. UV-vis absorption spectra of Glass/ITO/PTAA and Glass/ITO/PTAA/MoS₂.

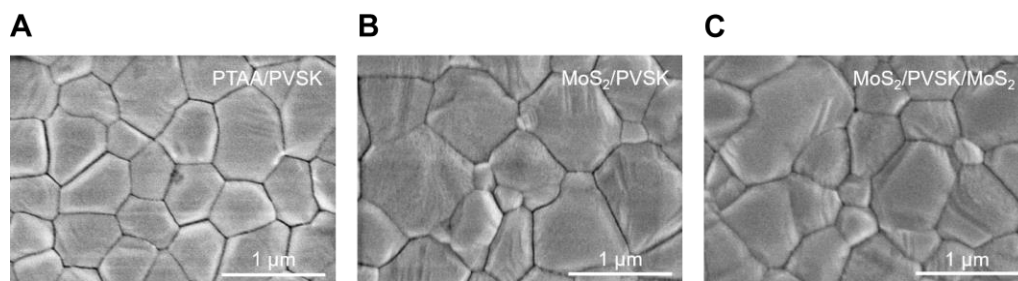


Fig. S5. (A to C) SEM images of PTAA/perovskite (A), PTAA/MoS₂/perovskite (B), and PTAA/MoS₂/perovskite/MoS₂ (C) films.

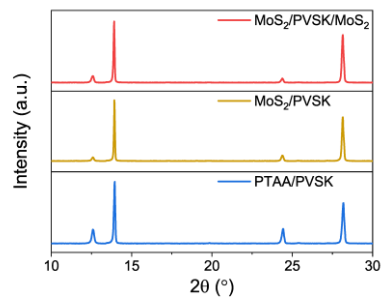


Fig. S6. XRD patterns of PTAA/perovskite, PTAA/ MoS_2 /perovskite, and PTAA/ MoS_2 /perovskite/ MoS_2 films.

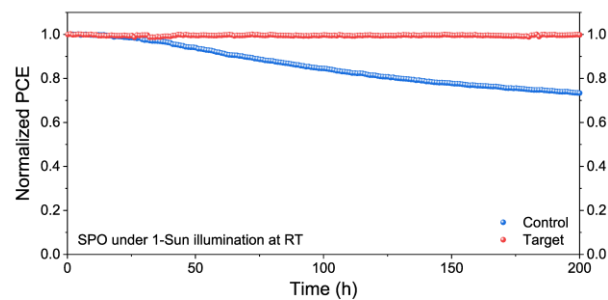


Fig. S7. SPO tracking for devices under 1-Sun illumination at room temperature in a nitrogen atmosphere.

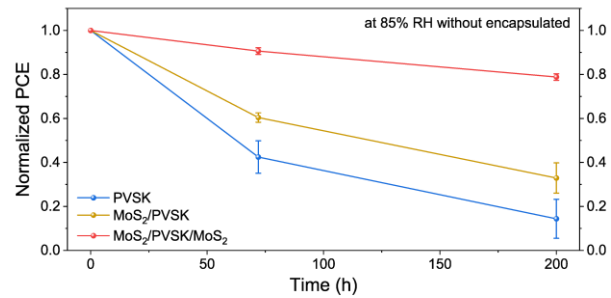


Fig. S8. Humidity stability tests for unencapsulated devices at RT and 85% RH.

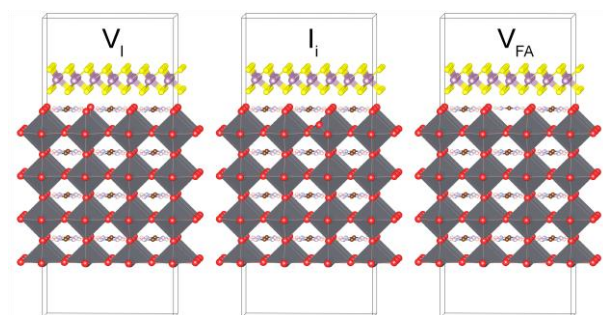


Fig. S9. Schematic illustration of MoS_2 passivated (001) FAI-terminated FAPbI_3 surface with V_I , I_i and V_{FA} .

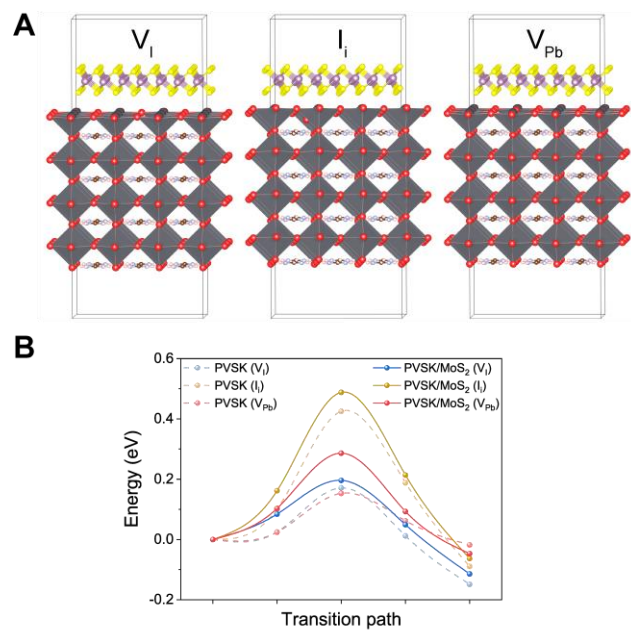


Fig. S10. Theoretical simulation. (A) Schematic illustration of MoS₂ passivated PbI₂-terminated FAPbI₃ surface with V_I, I_i and V_{Pb}. (B) Energy curves of phase transition for the perovskite without and with MoS₂ passivation.

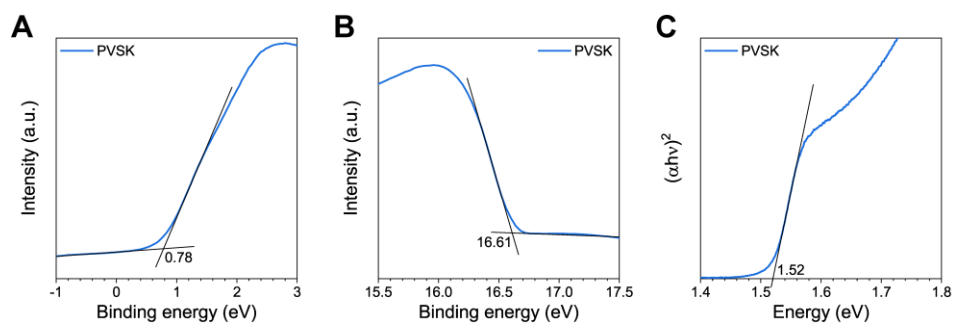


Fig. S11. Energy band characterization of perovskite film. (**A** and **B**) UPS spectrum of perovskite film. (**C**) Tauc plot of perovskite film from UV-vis absorption spectrum.

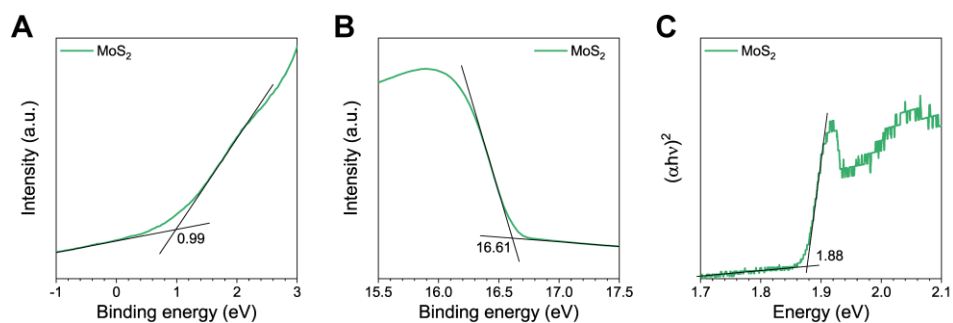


Fig. S12. Energy band characterization of monolayer MoS₂ film. (A and B) UPS spectrum of monolayer MoS₂ film. (C) Tauc plot of monolayer MoS₂ film from UV-vis absorption spectrum.

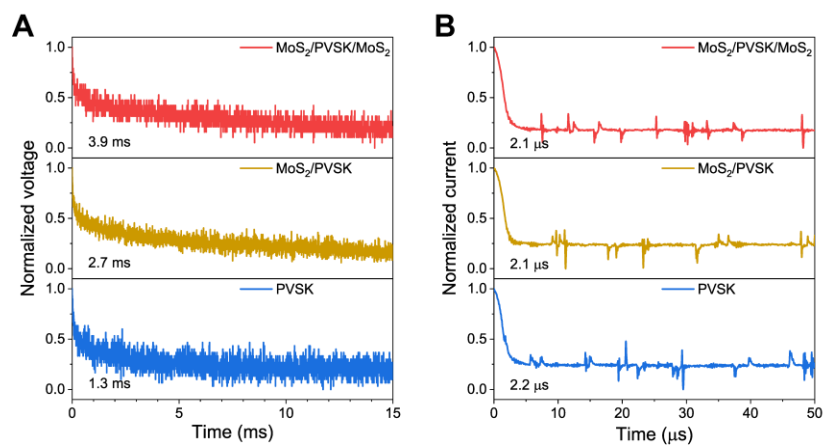


Fig. S13. (A and B) TPV (A) and TPC (B) decay curves of PSCs with different configurations.

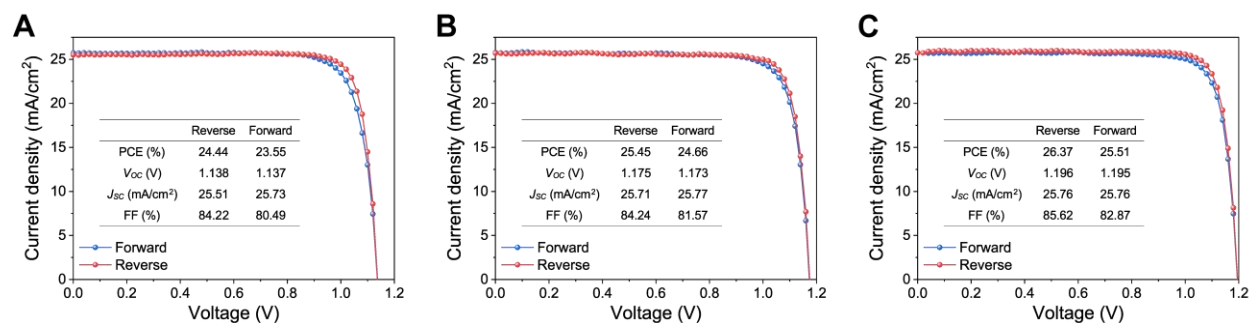


Fig. S14. (A to C) *J-V* curves of solar cells with architectures of ITO/PTAA/FAPbI₃/C₆₀/BCP/Ag (A), ITO/PTAA/MoS₂/FAPbI₃/C₆₀/BCP/Ag (B), and ITO/PTAA/MoS₂/FAPbI₃/MoS₂/C₆₀/BCP/Ag (C), respectively.

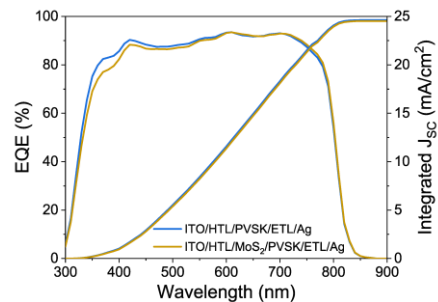


Fig. S15. EQE spectra together with integrated J_{SC} of ITO/PTAA/FAPbI₃/C₆₀/BCP/Ag and ITO/PTAA/MoS₂/FAPbI₃/C₆₀/BCP/Ag. Although MoS₂ exhibits absorption in the range of 350-680 nm, the measured current loss of the device is minimal due to improved carrier collection and possible energy transfer.

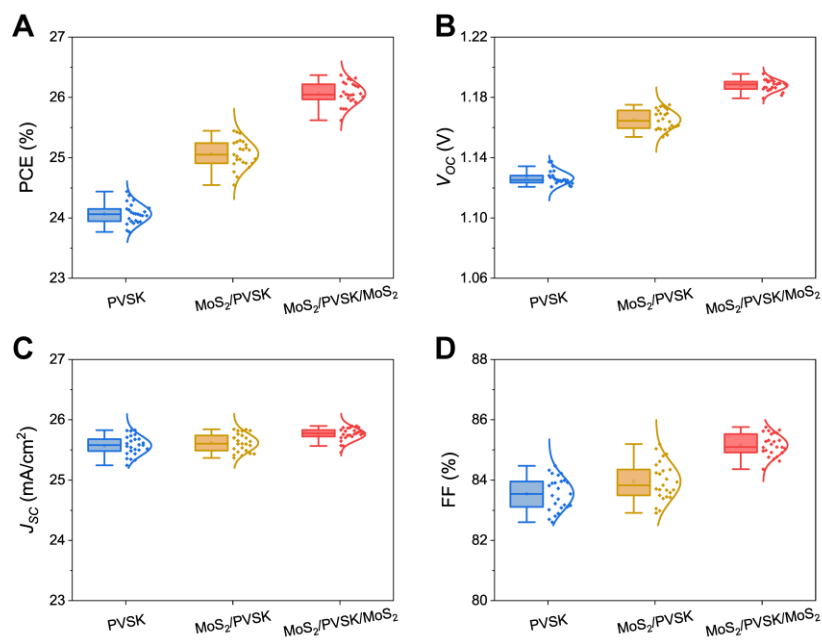


Fig. S16. Photovoltaic parameters of FAPbI₃-based inverted solar cells with different configurations.

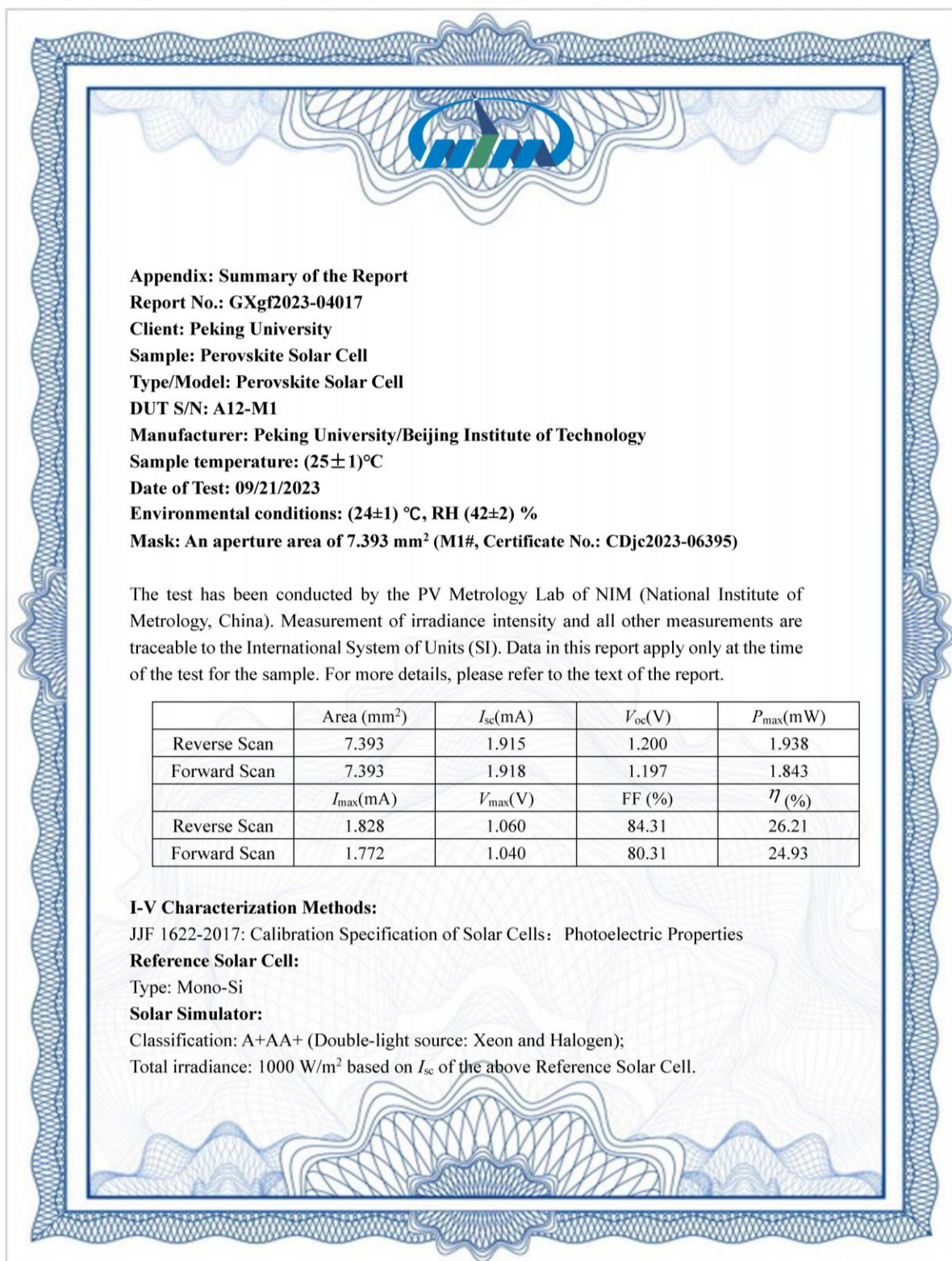


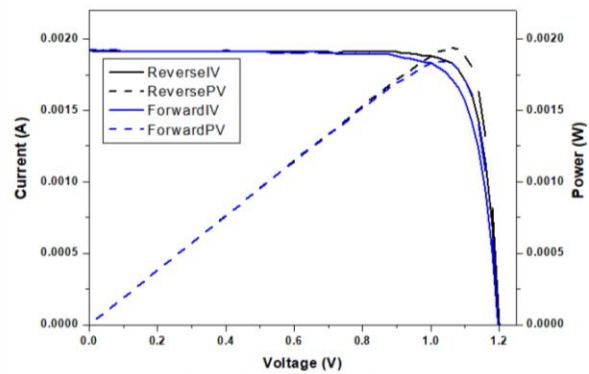
Fig. S17. Certificated results from the National Institute of Metrology, China (NIM, China). The forward scan (reverse scan) is performed from −0.10 V to 1.24 V (1.24 V to −0.1 V) with ~70 mV/s.



Report No.: GXgf2023-04017

DUT S/N: A12-M1

I-V Characteristic curves and parameters (Reverse & Forward Scan)



Steady-state scanning curve ($V_{\text{bias}} = 1.050$ V, $t_{\text{scanning}} = 300$ s, $P_{\text{start}} = 1.925$ mW and $P_{\text{end}} = 1.914$ mW)

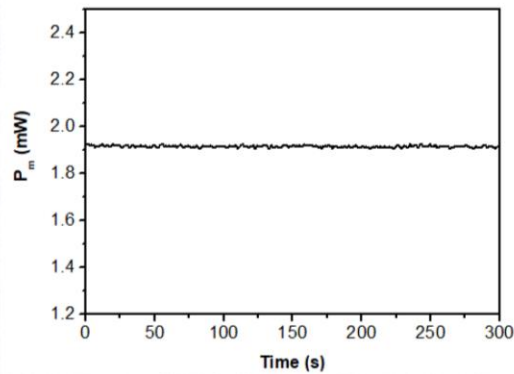


Fig. S18. Certificated results from the National Institute of Metrology, China (NIM, China). The steady-state scanning is performed for 300 s with one read per second under 1.050 V.

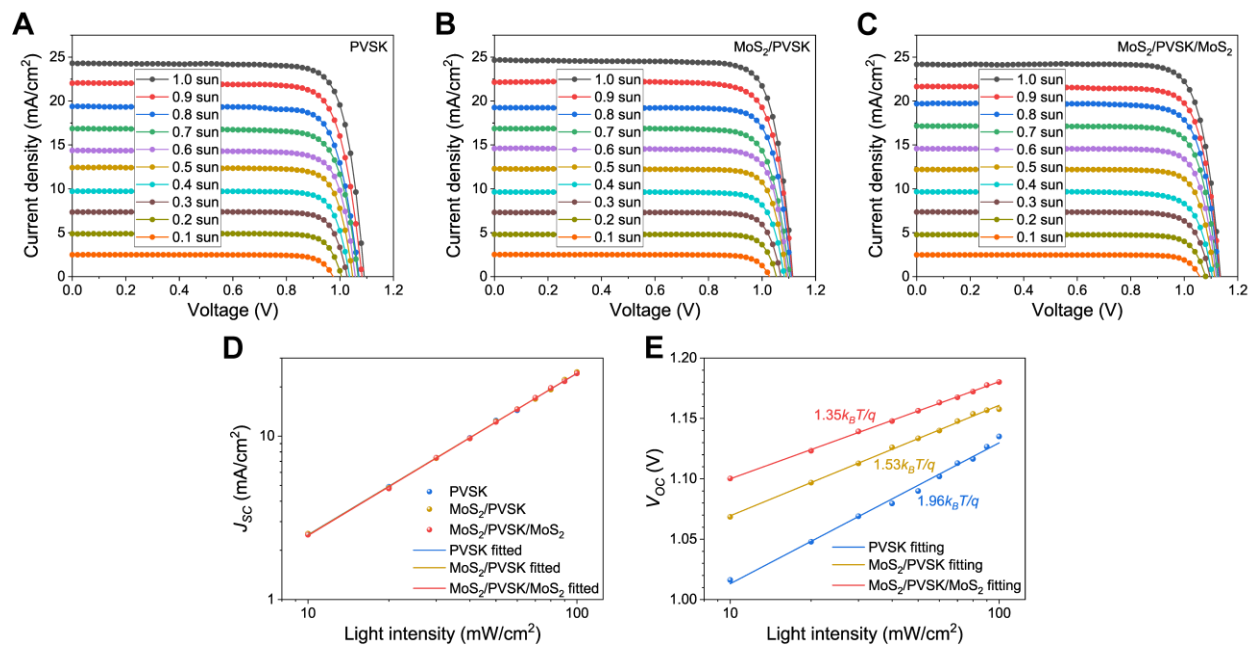


Fig. S19. The response of solar cells under different light intensities. **(A)** J - V curves of the control device under different light intensity ranging from 0.1 Sun to 1 Sun. **(B)** J - V curves of the device with MoS₂/PVSK under different light intensity ranging from 0.1 Sun to 1 Sun. **(C)** J - V curves of the device with MoS₂/PVSK/MoS₂ under different light intensity ranging from 0.1 Sun to 1 Sun. **(D and E)** Relationship between J_{sc} (D) and V_{oc} (E) vs light intensity for these three devices.

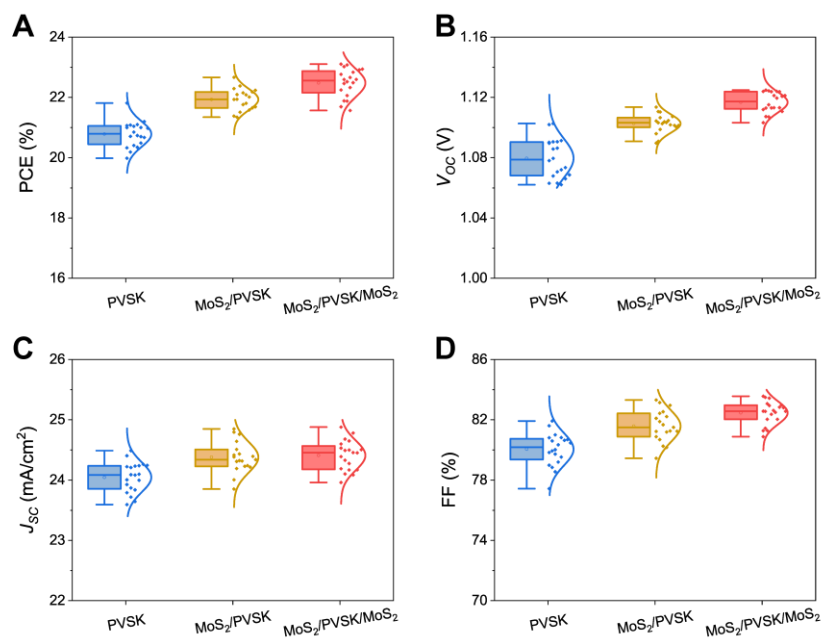


Fig. S20. Photovoltaic parameters of $\text{FA}_{0.93}\text{Cs}_{0.07}\text{PbI}_3$ -based inverted solar cells with different configurations.

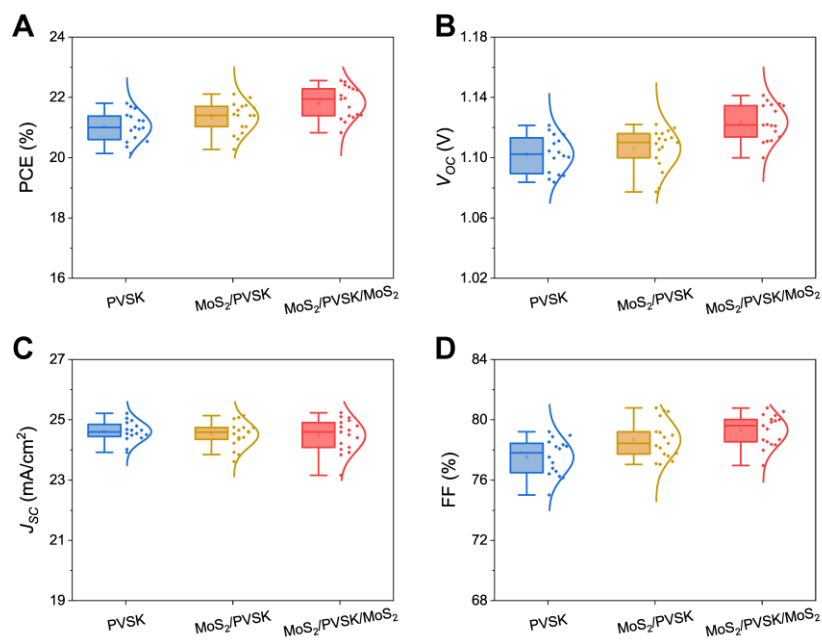


Fig. S21. Photovoltaic parameters of FAPbI₃-based regular solar cells with different configurations.

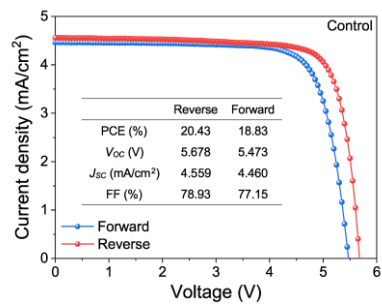


Fig. S22. *J*-*V* curve of the best-performing minimodule with the architecture of ITO/PTAA/FAPbI₃/C₆₀/SnO₂/Cu.

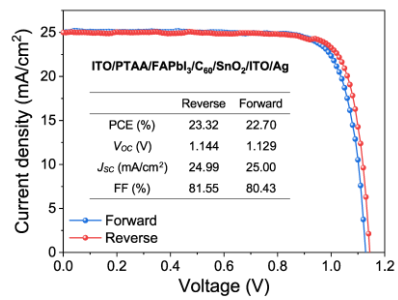


Fig. S23. *J*-*V* curves of the PSC with the architecture of ITO/PTAA/FAPbI₃/C₆₀/SnO₂/ITO/Ag.

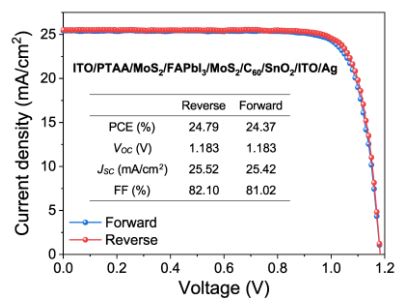


Fig. S24. *J-V* curves of the PSC with the architecture of ITO/PTAA/MoS₂/FAPbI₃/MoS₂/C₆₀/SnO₂/ITO/Ag.

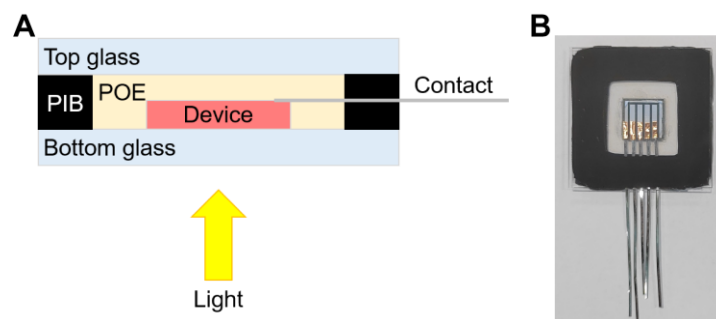


Fig. S25. Encapsulation. **(A)** Schematic representation of the materials and layers for the encapsulation. **(B)** Photograph of an encapsulated device from the rear contact.

Damp-heat chamber



Inside chamber



Fig. S26. Photographs of a damp-heat chamber in our lab.

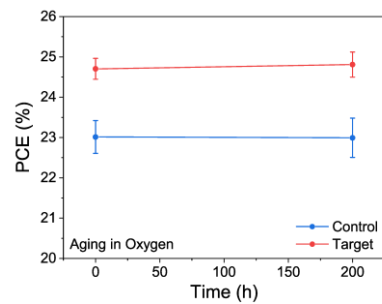


Fig. S27. Stability of unencapsulated devices under dark storage in an oxygen atmosphere.

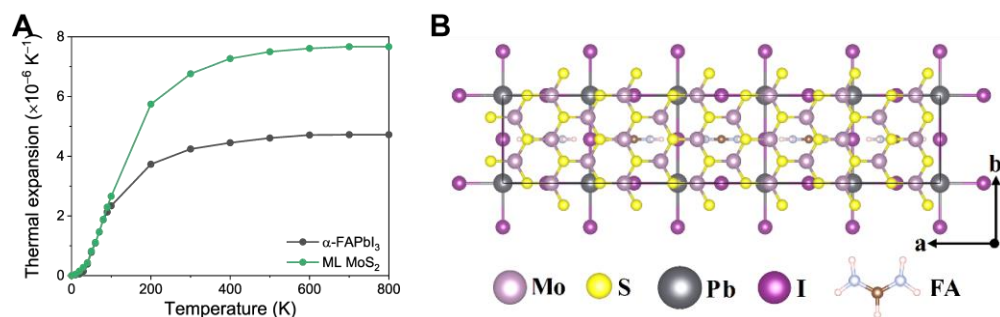


Fig. S28. (A) The thermal expansion coefficients of monolayer MoS_2 and FAPbI_3 under different temperature obtained by the first-principles simulations. (B) Top view of $\text{MoS}_2/\text{FAPbI}_3/\text{MoS}_2$ system.

Supplementary note 1

We have calculated the linear thermal expansion coefficients (α) of monolayer (ML) MoS₂ and FAPbI₃ using the first-principles simulations. The detailed linear thermal expansion coefficients are listed in fig. S27A and table S2. The linear thermal expansion coefficient of FAPbI₃ is slightly lower than that of monolayer MoS₂, which is consistent with previous reports. According to the linear thermal expansion coefficients, the lattice constants ($a=b$) of ML MoS₂ and FAPbI₃ under different temperatures are calculated, as listed in table S2. Consequently, the interfacial lattice constant mismatch between ML MoS₂ and FAPbI₃ of the ML MoS₂/FAPbI₃ system under the different temperature is obtained, as illustrated in table S2. Since ML MoS₂ shows the higher thermal expansion coefficient and thinner thickness than those of FAPbI₃, the interfacial stress is theoretically appeared at the ML MoS₂ of ML MoS₂/FAPbI₃ system. Thus, the interfacial stress is applied to the ML MoS₂ in the ML MoS₂/ FAPbI₃ system model. Such model can be composed by the $6\sqrt{3} \times 2$ supercell of ML MoS₂ and the 5×1 supercell of FAPbI₃ (100) surface, as displayed in fig. S27B. The negative and positive stress represent the compression and tension, respectively. It is obvious that ML MoS₂ undergoes the slightly enlarged tension as temperature varies from room temperature to -40 °C, while ML MoS₂ undergoes the slightly enlarged compression as temperature varies from room temperature to $+85$ °C. The strain induced by the different thermal expansion coefficients are small and less than 0.03 %.

To evaluate the interfacial adhesion of ML MoS₂/FAPbI₃ system, the interfacial binding energy is calculated by $E_b = (E_{\text{system}} - E_{\text{MoS}_2} - E_{\text{FAPbI}_3})/S$, where E_{system} , E_{MoS_2} , and E_{FAPbI_3} represent the total energies of ML MoS₂/FAPbI₃ system, isolated ML MoS₂, isolated FAPbI₃ surface, respectively. S indicates the interfacial area. The interfacial binding energies are listed in table S2. It can be found that these ML MoS₂/FAPbI₃ models exhibit similar interfacial binding energies, suggesting that the small interfacial stress has a negligible effect on the interfacial adhesion. In other words, the temperature region (-40 to $+85$ °C) has a negligible effect on the interfacial adhesion.

Thermal cycling chamber



Inside chamber



Fig. S29. Photographs of a thermal cycling chamber in our lab.

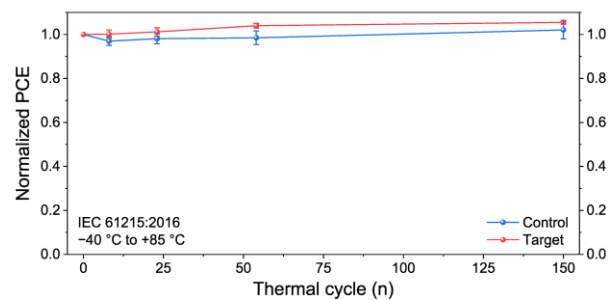


Fig. S30. Thermal cycling (between $-40\text{ }^{\circ}\text{C}$ and $+85\text{ }^{\circ}\text{C}$) tests for encapsulated control and target devices.

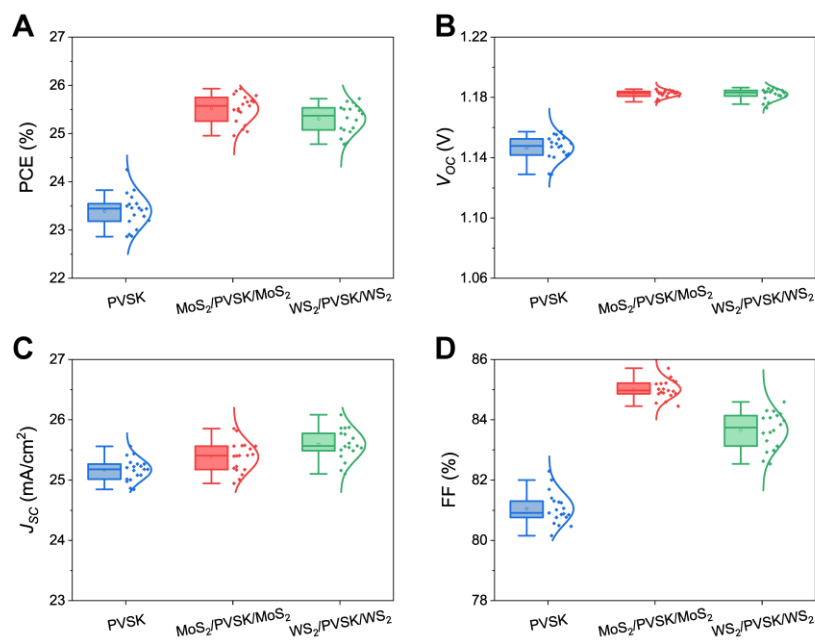


Fig. S31. Photovoltaic parameters of solar cells with architectures of ITO/PTAA/FAPbI₃/C₆₀/SnO₂/Ag, ITO/PTAA/MoS₂/FAPbI₃/MoS₂/C₆₀/SnO₂/Ag, and ITO/PTAA/WS₂/FAPbI₃/WS₂/C₆₀/SnO₂/Ag.

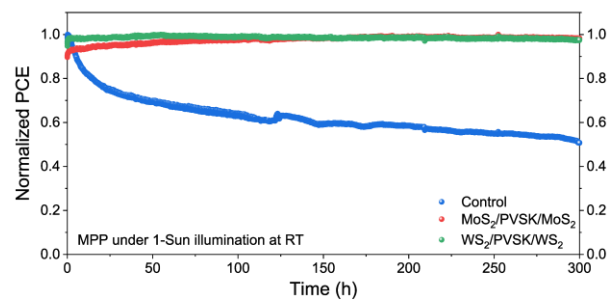


Fig. S32. MPP tracking for encapsulated devices under 1-Sun illumination at room temperature in the ambient atmosphere.

Table S1. Photovoltaic parameters of solar cells fabricated with different configuration.

Device		V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)
PVSK	Best	1.138	25.51	84.22	24.44
	Average	1.126±0.005	25.57±0.16	83.55±0.55	24.07±0.17
MoS ₂ /PVSK	Best	1.175	25.71	84.24	25.45
	Average	1.165±0.007	25.62±0.15	83.96±0.64	25.06±0.23
MoS ₂ /PVSK/MoS ₂	Best	1.196	25.76	85.62	26.37
	Average	1.188±0.004	25.76±0.11	85.16±0.37	26.06±0.19

Average device characteristics with standard deviation were obtained based on 25 cells for each set.

Table S2. Photovoltaic parameters of solar cells fabricated with different configuration.

T		α (10^{-6}K^{-1})		a = b (Å)		ϵ_{MoS_2} (%)	$\Delta\epsilon_{\text{MoS}_2}$ (%)	Binding energy (eV/Å ²)
°C	K	MoS ₂	FAPbI ₃	MoS ₂	FAPbI ₃	a&b	a&b	
-40	233	5.9803	3.8537	3.1670	6.3596	-1.5496	0.0258	-0.0388
0	273	6.4446	4.0859	3.1681	6.3610	-1.5625	0.0130	-0.0389
27	300	6.7600	4.2400	3.1690	6.3620	-1.5754	0	-0.0392
40	313	6.7637	4.2498	3.1693	6.3624	-1.5786	-0.0032	-0.0392
85	358	7.0018	4.3460	3.1705	6.3638	-1.5947	-0.0193	-0.0390

Movie S1. Transfer of monolayer MoS₂ film.